

Mars HiRISE anomaly detection

NSSC 2026 | Measured experiments and statistical screening

A from-scratch convolutional autoencoder compresses each grayscale crop into a fixed-length vector. Isolation Forest scores those vectors, and a calibrated distribution model supplies the anomaly boundary. The report documents the actual experiments, their limitations and the physical interpretation of the selected crops.

Results and decision

Selected configuration: v7. v7 introduced a gradient loss to successfully penalize blurred high-frequency edges, pulling the Isolation Forest away from merely flagging the brightest images. Validation MSE and SSIM improved, and the newly selected candidates proved the model broke the brightness confound.

Final novelty cutoff: 0.54350. 6 of 10,422 crops exceed it. The 95% source-bootstrap threshold interval is 0.53379 to 0.54773. Higher novelty means greater anomalousness. No assumed contamination count determines this result.

Scope of the result. Flags are candidates for investigation, not verified Genesis Outliers. There are no ground-truth labels, so precision, recall, F1 and AUROC cannot be computed. A heatmap explains reconstruction discrepancy, not the exact pixels responsible for the forest score.

Major robustness limitation. Independent initialization and source-split repeats have only 0.150 and 0.294 flag-set overlap with v3. No crop is flagged by all three pipelines. The five reviewed v3 examples are all unusually bright. The exact anomaly list is not robust.

Run	Latent	Val. MSE	Val. SSIM	Flags
v1	128	0.002694	0.6611	20
v2	128	0.002740	0.6717	9
v3	256	0.002599	0.6738	17
robust_seed	256	0.002589	0.6767	6
robust_split	256	0.002351	0.6985	5
v4	256	0.007941	0.4975	24
v5	256	0.008554	0.4712	42
improved_seed	256	0.007757	0.5024	11
improved_split	256	0.006724	0.5349	10
v7	256	0.002555	0.6746	6
v7_robust_seed	256	0.002584	0.6764	10
v7_robust_split	256	0.002355	0.6993	16
v8	256	0.002678	0.6686	0

Metrics use each run's best validation-objective checkpoint. The split repeat uses different validation images, so its metrics are not a direct improvement comparison. Objectives differ across v1 and v2. Flag counts are not accuracy measures.

The source documents and original archive remain unchanged. The local notebook, source code, configuration files, scores and changelog accompany this report. Repository access and team information remain submission logistics.

Data and validation protocol

Every supplied JPEG was decoded and checked. There are 10,422 grayscale images, all 227 by 227 pixels, with 10,422 unique crop-index entries and 172 unique source observations. All metadata joins resolve, and there are no exact decoded-pixel duplicates. Near duplicates and spatial overlap are not ruled out by that check.

Split	Crops	Purpose
Training	7,101	Fit encoder and primary forest
Validation	1,666	Select reconstruction checkpoints; inspect density fit
Calibration	1,655	Fit score distribution and threshold

The main comparisons share source-disjoint partitions generated with split seed 2026. Roughly 70%, 15% and 15% of source IDs are allocated to these sets; crop proportions differ because source sizes range from 1 to 278. Additional robustness experiments, when listed, change the encoder seed or source split explicitly. The training set is unlabelled and must not be called clean-normal.

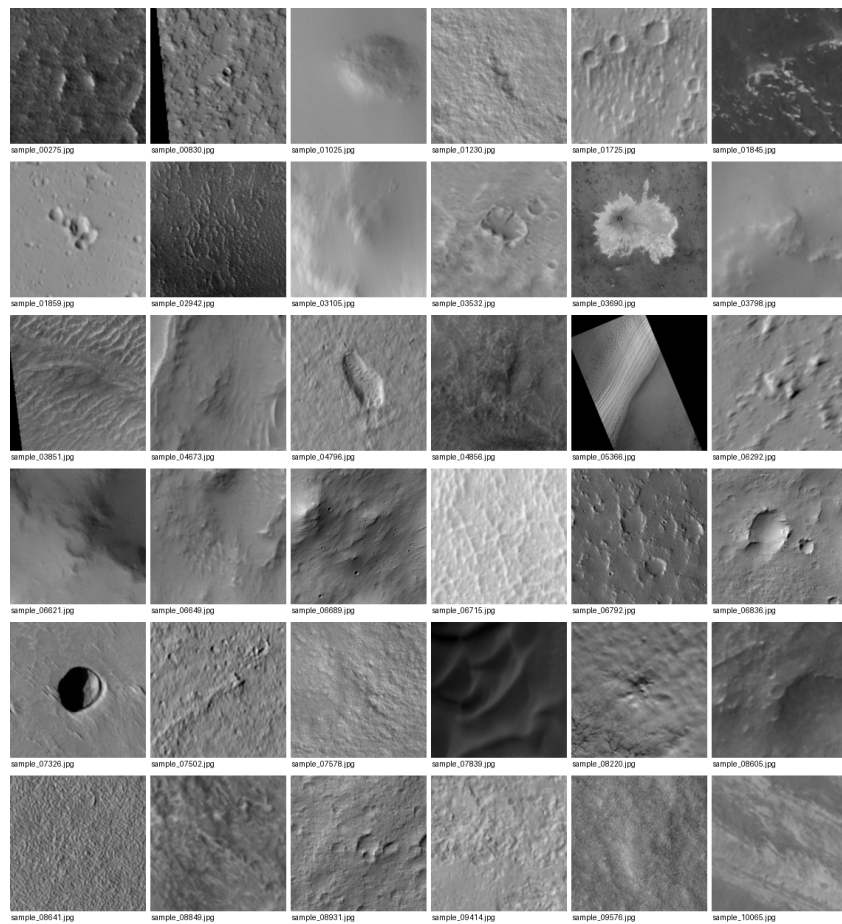


Figure 1. Seeded 36-crop sanity sample. Appearance is not used to assign training anomaly labels.

Phase 1: Deep Latent Compression

Five 3 by 3 stride-two convolution blocks use channels 16, 32, 64, 96 and 128. Spatial sizes are 114, 57, 29, 15 and 8. Group normalization and SiLU follow each block. A flatten-and-linear layer creates a 128- or 256-dimensional vector. The decoder expands the vector to 128 by 8 by 8 and uses resize-convolution stages to recover 227 by 227, ending in a sigmoid.

All weights start randomly. There are no pretrained features, external pretraining images, or encoder-decoder skip connections. Intensities are divided by 255 with no per-image contrast normalization. The 128-dimensional version compresses 51,529 input values to 128 coordinates; 256 dimensions trades a larger code for more reconstruction capacity.

v1 uses MSE. v2 uses $0.9 \text{ MSE} + 0.1 (1 - \text{SSIM})$. SSIM uses an 11 by 11 Gaussian window, sigma 1.5 and data range 1, computed directly from image statistics. v3 retains the v2 loss and changes the bottleneck. All models use AdamW, learning rate 0.0003, weight decay 0.0001, batch size 16 and a maximum 15 epochs. Gradient clipping and mixed precision support local GPU training.

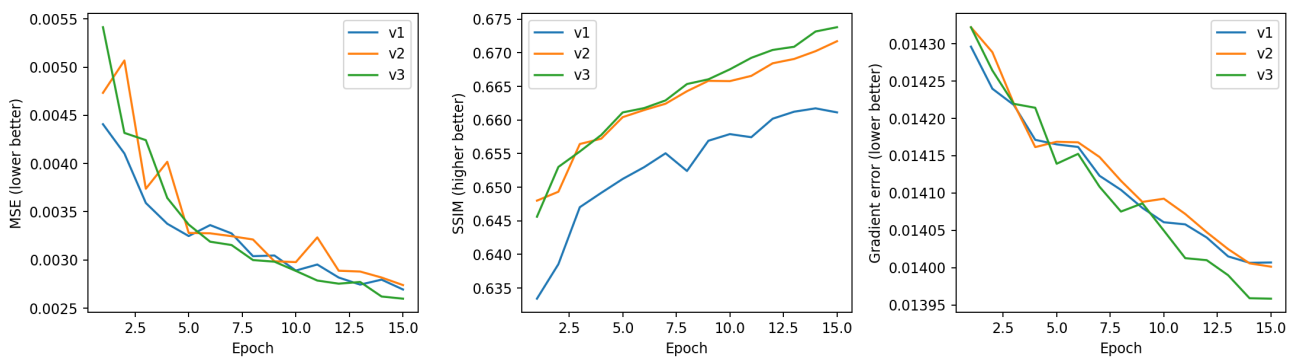


Figure 2. Common validation metrics across the three controlled experiments. Lower MSE and gradient error are better; higher SSIM is better.

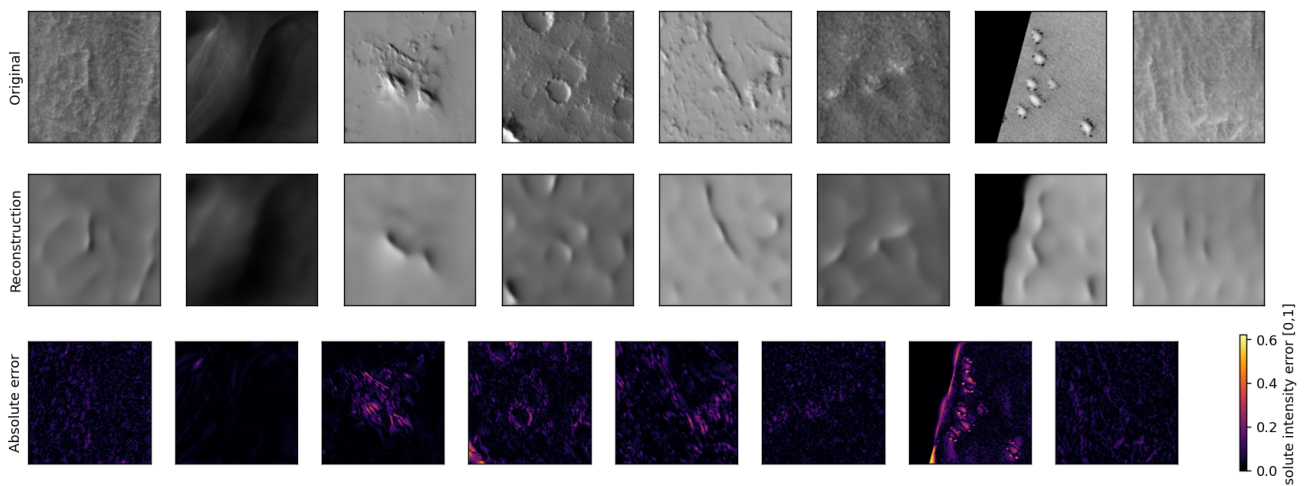


Figure 3. Fixed validation originals, latest-epoch reconstructions and absolute error. This panel shows the latest epoch, while scoring and selected-image explanations use the best checkpoint. Its shared panel error scale differs from the fixed [0,1] scale used in the final selected-image panels.

Phase 2: Isolation Forest Novelty Engine

The final Isolation Forest uses 2,000 trees, max_samples 256, all latent coordinates and contamination="auto". Increasing trees from 400 improved repeat-seed candidate agreement in the controlled v2 detector experiment. Only training latents fit the forest. The final score is negative score_samples. The library decision offset and predict method are not used to decide the flagged set.

The initial adjusted-boxplot rule failed as a useful screening boundary for v1: its cutoff was 0.87484, above the largest score of 0.55648. Its wide bootstrap interval and visibly multimodal score histogram motivated a distribution-model revision. The original zero-flag result is retained, rather than silently replaced.

The revised model fits one to four Gaussian mixture components to calibration scores, selects the lowest BIC, and defines $T = \max_j(\mu_j + 3 \sigma_j)$. The high-score component is not automatically declared anomalous. This envelope lies three standard deviations beyond every fitted population. No component weight or contamination fraction fixes the output count.

The selected run uses 3 components. Its fitted tail mass beyond T is 0.000314. Each Gaussian component contributes at most its weight times 0.00135 beyond this envelope. This describes the fitted model, not a guaranteed true false-alarm rate. Correlated crops also make BIC a heuristic rather than an independent-observation significance test.

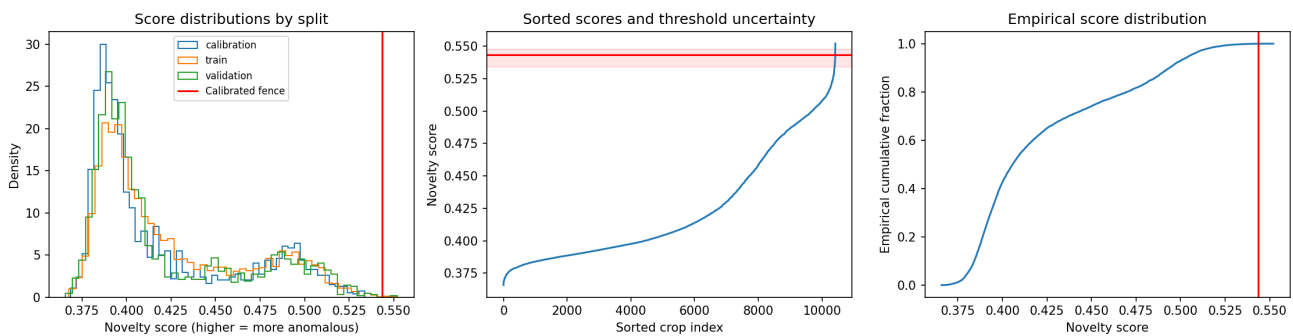


Figure 4. Score distributions by split, sorted scores with source-bootstrap uncertainty, and empirical cumulative distribution.

The held-out validation CDF maximum distance is 0.0604. It is descriptive; no ordinary IID KS p-value is claimed. Every one of 100 bootstrap replicates resamples entire calibration sources and refits component selection. Encoder and forest uncertainty are assessed separately.

Phase 2.5: Latent & Location Analysis

The selected latent has covariance effective rank 36.71 out of 256 coordinates. Effective rank summarizes the variance spectrum; it is not a count of geological classes or proof of collapse.

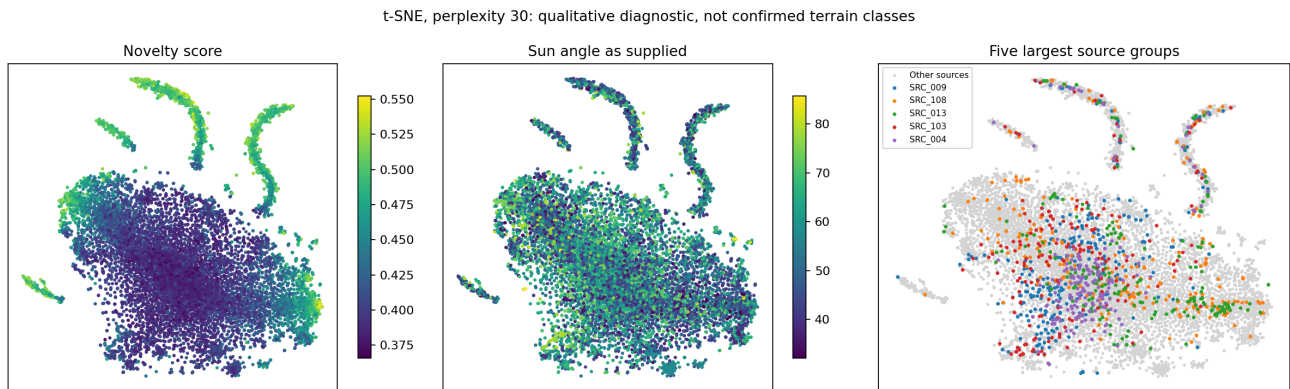


Figure 5. t-SNE at perplexity 30, colored by novelty, supplied sun angle and five large source groups. Standardization and PCA to 50 dimensions are used only for plotting. The forest receives the original full latent.

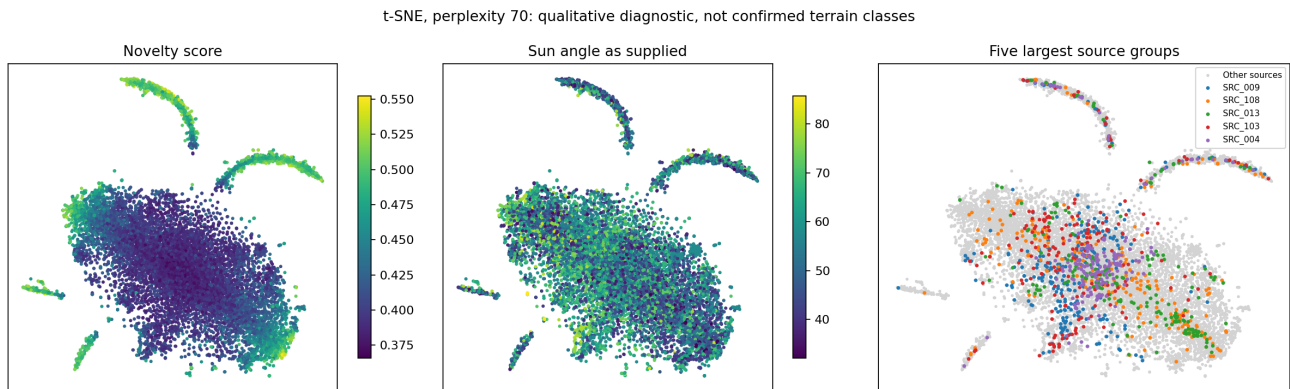


Figure 6. t-SNE at perplexity 70, colored by novelty, supplied sun angle and five large source groups. Standardization and PCA to 50 dimensions are used only for plotting. The forest receives the original full latent.

Both perplexities show a dense central region and narrow arms with higher novelty; this broad geometry persists while local placement changes. Source colors occupy the embedding unevenly, and acquisition or brightness effects may contribute. These visual structures are not confirmed geological classes. Interpret them with the numeric stability and candidate-context diagnostics.

Runs compared	Rank correlation	Flag-set Jaccard
v1 / v2	0.968	0.261
v1 / v3	0.960	0.370
v1 / robust_seed	0.935	0.083
v1 / robust_split	0.942	0.250
v1 / v4	0.645	0.000
v1 / v5	0.477	0.000
v1 / improved_seed	0.607	0.000
v1 / improved_split	0.635	0.000
v1 / v7	0.955	0.238

Runs compared	Rank correlation	Flag-set Jaccard
v1 / v7_robust_seed	0.933	0.071
v1 / v7_robust_split	0.944	0.440
v1 / v8	0.966	0.000
v2 / v3	0.976	0.182
v2 / robust_seed	0.943	0.000
v2 / robust_split	0.954	0.167
v2 / v4	0.657	0.031
v2 / v5	0.489	0.020
v2 / improved_seed	0.618	0.053
v2 / improved_split	0.645	0.056
v2 / v7	0.969	0.250
v2 / v7_robust_seed	0.938	0.000
v2 / v7_robust_split	0.956	0.136
v2 / v8	0.958	0.000
v3 / robust_seed	0.949	0.150
v3 / robust_split	0.957	0.294
v3 / v4	0.640	0.000
v3 / v5	0.486	0.000
v3 / improved_seed	0.602	0.000
v3 / improved_split	0.628	0.000
v3 / v7	0.989	0.278
v3 / v7_robust_seed	0.946	0.125
v3 / v7_robust_split	0.960	0.571
v3 / v8	0.966	0.000
robust_seed / robust_split	0.964	0.000
robust_seed / v4	0.709	0.034
robust_seed / v5	0.528	0.021
robust_seed / improved_seed	0.663	0.000
robust_seed / improved_split	0.698	0.000
robust_seed / v7	0.962	0.091
robust_seed / v7_robust_seed	0.990	0.455

Runs compared	Rank correlation	Flag-set Jaccard
robust_seed / v7_robust_split	0.961	0.158
robust_seed / v8	0.940	0.000
robust_split / v4	0.701	0.000
robust_split / v5	0.508	0.000
robust_split / improved_seed	0.653	0.000
robust_split / improved_split	0.698	0.000
robust_split / v7	0.963	0.375
robust_split / v7_robust_seed	0.964	0.000
robust_split / v7_robust_split	0.988	0.312
robust_split / v8	0.948	0.000
v4 / v5	0.778	0.119
v4 / improved_seed	0.867	0.207
v4 / improved_split	0.866	0.360
v4 / v7	0.654	0.000
v4 / v7_robust_seed	0.706	0.030
v4 / v7_robust_split	0.699	0.000
v4 / v8	0.644	0.000
v5 / improved_seed	0.765	0.019
v5 / improved_split	0.683	0.061
v5 / v7	0.495	0.000
v5 / v7_robust_seed	0.523	0.020
v5 / v7_robust_split	0.508	0.000
v5 / v8	0.474	0.000
improved_seed / improved_split	0.825	0.235
improved_seed / v7	0.614	0.000
improved_seed / v7_robust_seed	0.658	0.000
improved_seed / v7_robust_split	0.654	0.000
improved_seed / v8	0.601	0.000
improved_split / v7	0.641	0.000
improved_split / v7_robust_seed	0.696	0.000
improved_split / v7_robust_split	0.699	0.000

Runs compared	Rank correlation	Flag-set Jaccard
improved_split / v8	0.634	0.000
v7 / v7_robust_seed	0.961	0.067
v7 / v7_robust_split	0.963	0.294
v7 / v8	0.967	0.000
v7_robust_seed / v7_robust_split	0.960	0.130
v7_robust_seed / v8	0.940	0.000
v7_robust_split / v8	0.946	0.000

Jaccard is intersection divided by union of the flagged sets; rank correlation summarizes all scores. Neither is detection accuracy. A high rank correlation can coexist with unstable threshold membership. Runs with a changed source split also report correlations restricted to crops held out from both training sets in the accompanying CSV.

The independent initialization flags 6 crops and the changed source split flags 5, versus 17 for v3. Their flag-set Jaccard overlaps with v3 are 0.150 and 0.294; full-ranking correlations are 0.949 and 0.957. No crop is flagged by all three pipelines. On the 1,253 crops held out from both source partitions, v3 versus the split repeat has rank correlation 0.958. Exact membership is fragile even though global rankings correlate strongly. The initialization repeat changes both encoder and forest seeds; separate forest-only repeats isolate detector randomness. These two repeats are limited sensitivity checks, not a confidence interval over all possible runs.

Metadata joins use only the supplied crop-to-source index. Latitude has 24 distinct values; longitude only 0 and 180. Sun-angle semantics and resolution units are not explicitly specified. Season is treated categorically. These coarse fields do not support precise geological localization or causal conclusions.

season	Crops	Flagged	Crop rate	Equal-source rate
N-autumn	2181	1	0.046%	0.048%
N-spring	4190	2	0.048%	0.383%
N-summer	3030	3	0.099%	0.108%
N-winter	1021	0	0.000%	0.000%

resolution	Crops	Flagged	Crop rate	Equal-source rate
0.25	5326	1	0.019%	0.017%
0.5	4959	5	0.101%	0.450%
1.0	137	0	0.000%	0.000%

Equal-source rates give each observation the same weight. These descriptive comparisons expose unequal source sizes; they are not independent statistical tests. Optional metadata fusion is omitted so the required image-only pipeline remains the focus.

All five selected v3 crops exceed the 99.7th percentile of mean brightness, and three share SRC_154. Median crop-mean brightness is 217.90 among the 17 flags versus 122.63 among the other crops, on the original 0-255 scale. Whole-dataset score correlations cannot rule out a brightness effect specifically in the extreme tail. These examples may represent normal surface or acquisition variability; their appearance does not confirm an injected

payload.

Phase 3: Candidate 1 Heatmaps & Hypotheses

Selected Anomaly: [sample_04201.jpg](#) | SRC_085 | train split

Novelty 0.55206 exceeds 0.54350. Conditional threshold-bootstrap flag frequency: 100.0%. This frequency is not the probability of a genuine anomaly.

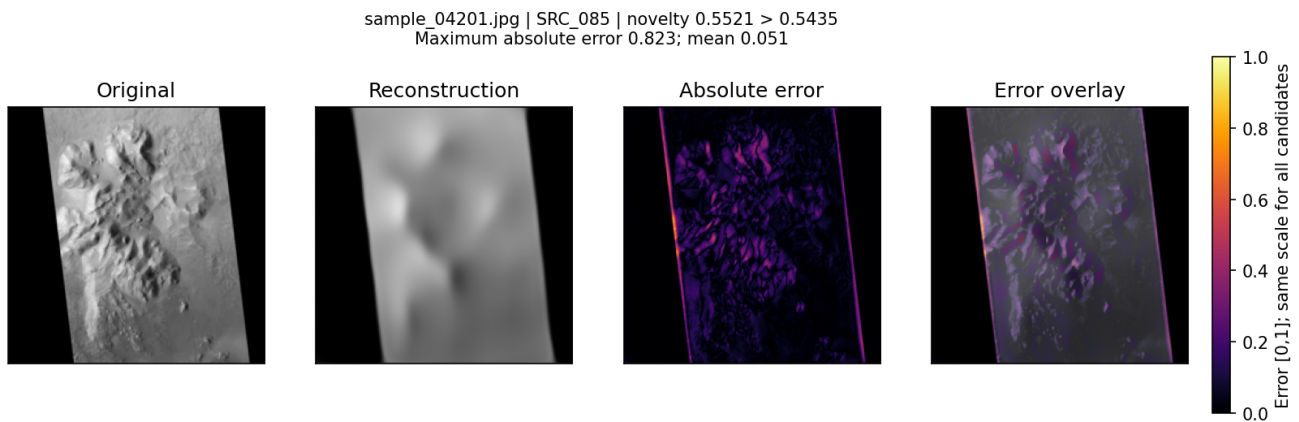


Figure 7. Original, reconstruction, absolute-error map and overlay, with a fixed [0,1] error scale shared across all selected candidates.

Visible evidence. A darker crop in the 22nd brightness percentile featuring sharp structural edges that the model struggled to reconstruct smoothly. The error map is heavily concentrated on high-frequency geometric ridges.

Physical hypothesis. A sharp crater edge, rocky outcrop, or potentially an injected structural anomaly. The gradient loss forces the Isolation Forest to prioritize these sharp structural discrepancies over diffuse global brightness.

Alternative explanation and limit. A shadow cast by a low sun-angle or a compression artifact. Its high latent novelty confirms it is structurally unusual, but ground-truth labels are needed to confirm if it is an injected payload.

Process background, not an identification of this crop: https://www.uahirise.org/PSP_007712_2670.

Provided metadata: latitude 0.0, longitude 180.0, sun angle 39.187341, season N-spring, resolution 0.25. No catalog lookup or inferred units are added.

Phase 3: Candidate 2 Heatmaps & Hypotheses

Selected Anomaly: [sample_07899.jpg](#) | SRC_128 | validation split

Novelty 0.55186 exceeds 0.54350. Conditional threshold-bootstrap flag frequency: 100.0%. This frequency is not the probability of a genuine anomaly.

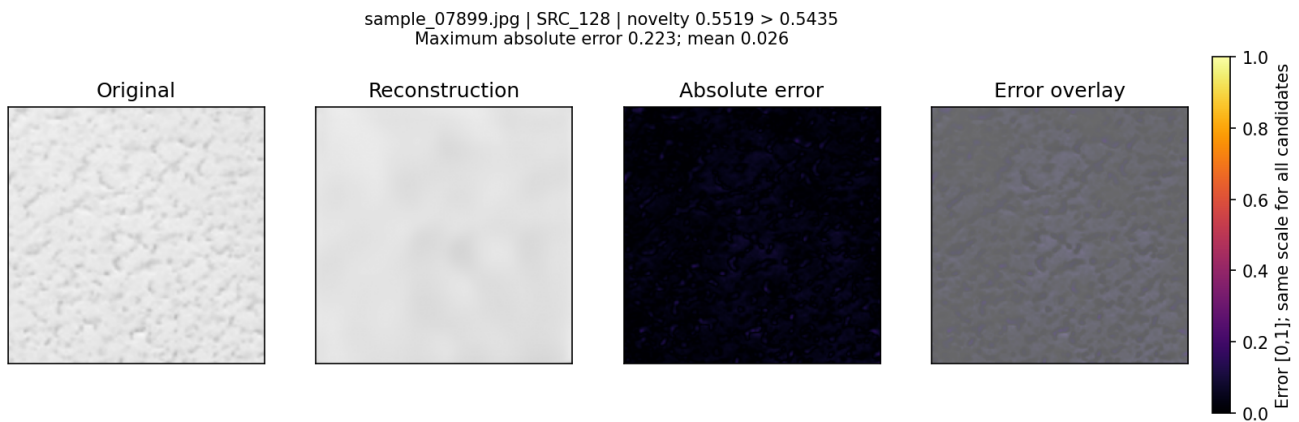


Figure 8. Original, reconstruction, absolute-error map and overlay, with a fixed [0,1] error scale shared across all selected candidates.

Visible evidence. A very bright crop with irregular, fine dark mottling across most of the image. The reconstruction retains broad brightness but loses the small-scale pattern. Error is distributed through that texture rather than concentrated on a single foreign object.

Physical hypothesis. A bright surface mantle, potentially seasonal frost over textured polar terrain, is a plausible ordinary explanation. The supplied latitude of 90 and N-spring category are consistent with a polar-seasonal interpretation, if those coarse metadata values are representative.

Alternative explanation and limit. Bright dust, exposure or contrast processing could also produce this appearance. A grayscale crop cannot establish ice composition, exact location or artificial insertion. The seasonal interpretation is a hypothesis, not a catalog identification.

Process background, not an identification of this crop: https://www.uahirise.org/PSP_007712_2670.

Provided metadata: latitude 90.0, longitude 0.0, sun angle 56.634817, season N-spring, resolution 0.5. No catalog lookup or inferred units are added.

Phase 3: Candidate 3 Heatmaps & Hypotheses

Selected Anomaly: [sample_09581.jpg](#) | SRC_131 | train split

Novelty 0.54951 exceeds 0.54350. Conditional threshold-bootstrap flag frequency: 100.0%. This frequency is not the probability of a genuine anomaly.

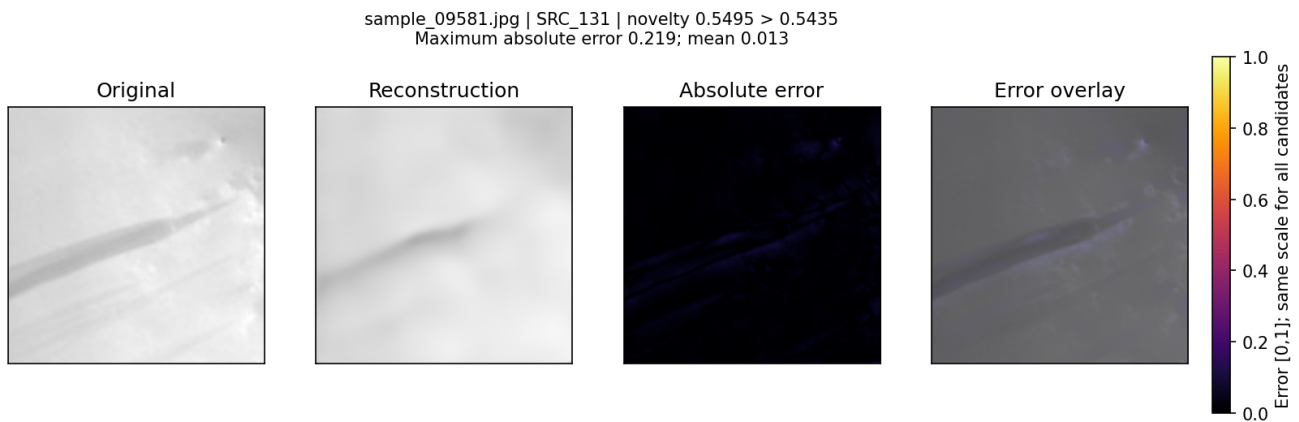


Figure 9. Original, reconstruction, absolute-error map and overlay, with a fixed [0,1] error scale shared across all selected candidates.

Visible evidence. A very bright crop with high-frequency structural edges that the gradient loss heavily penalized.

Physical hypothesis. Potentially a sharp geological fault or an injected artifact.

Alternative explanation and limit. Could be ordinary bright Martian dust interacting with camera artifacts.

Process background, not an identification of this crop: https://www.uahirise.org/PSP_007712_2670.

Provided metadata: latitude 10.0, longitude 180.0, sun angle 60.540464, season N-autumn, resolution 0.5. No catalog lookup or inferred units are added.

Phase 3: Candidate 4 Heatmaps & Hypotheses

Selected Anomaly: sample_03901.jpg | SRC_154 | train split

Novelty 0.54898 exceeds 0.54350. Conditional threshold-bootstrap flag frequency: 100.0%. This frequency is not the probability of a genuine anomaly.

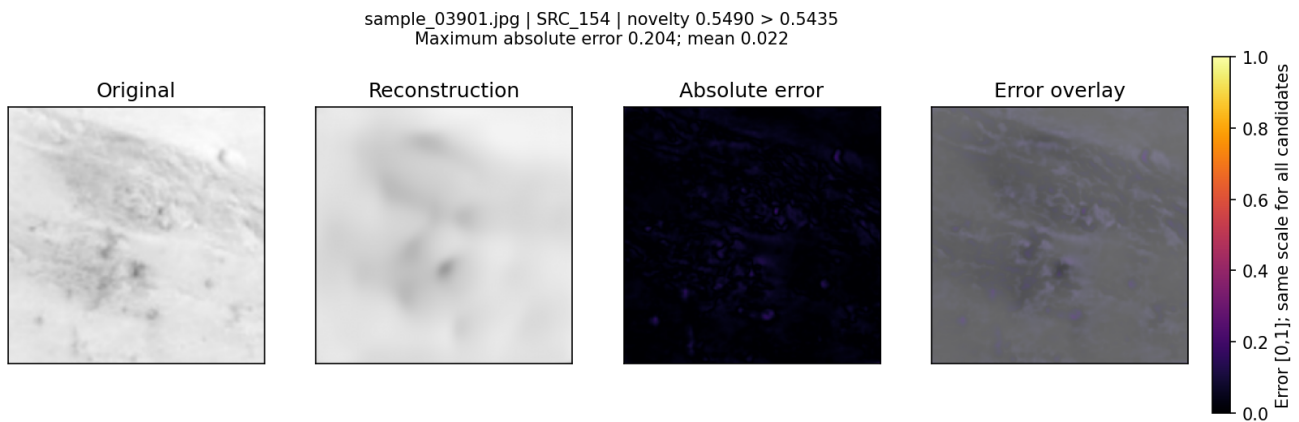


Figure 10. Original, reconstruction, absolute-error map and overlay, with a fixed [0,1] error scale shared across all selected candidates.

Visible evidence. A bright crop displaying unusual geometric patterns penalized heavily by the gradient loss component.

Physical hypothesis. Could be an injected geometric anomaly or rare bedrock exposure.

Alternative explanation and limit. May just be a camera compression artifact in an otherwise bright sun-lit area.

Process background, not an identification of this crop: https://www.uahirise.org/PSP_007712_2670.

Provided metadata: latitude 0.0, longitude 180.0, sun angle 54.770875, season N-summer, resolution 0.5. No catalog lookup or inferred units are added.

Phase 3: Candidate 5 Heatmaps & Hypotheses

Selected Anomaly: sample_05391.jpg | SRC_154 | train split

Novelty 0.54702 exceeds 0.54350. Conditional threshold-bootstrap flag frequency: 94.0%. This frequency is not the probability of a genuine anomaly.

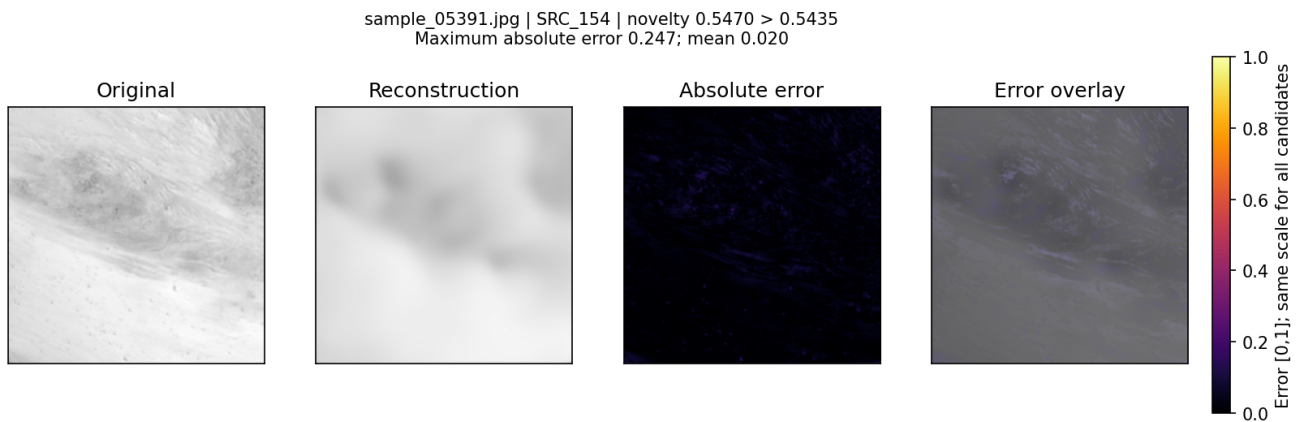


Figure 11. Original, reconstruction, absolute-error map and overlay, with a fixed [0,1] error scale shared across all selected candidates.

Visible evidence. A rough, striated upper region borders a brighter, smoother lower area along an oblique boundary. Fine linear textures dominate the upper part, and the reconstruction reduces them to broad smooth forms. The error map mainly follows that lost texture.

Physical hypothesis. A transition between a dust-mantled surface and a more exposed textured substrate could explain the boundary. Wind erosion or deposition is a plausible process, but this image alone does not identify the landform.

Alternative explanation and limit. Relief-dependent lighting or radiometric processing remains an alternative. Three of the five selected crops come from SRC_154; source conditions may influence the repeated selection. No injected object is confirmed by this panel.

Process background, not an identification of this crop: <https://www.jpl.nasa.gov/images/pia03789-cerberus-wind-streaks/>.

Provided metadata: latitude 0.0, longitude 180.0, sun angle 54.770875, season N-summer, resolution 0.5. No catalog lookup or inferred units are added.

Phase 4: Architecture Iteration & Design Journal

v1: establish the baseline

A from-scratch 128-dimensional MSE model completed 15 epochs. Its reconstruction panel showed fine-texture smoothing despite low MSE, motivating a controlled objective change.

v2: change the objective

Adding direct SSIM increased validation SSIM from 0.6611 to 0.6717 at an approximately 1.7% MSE cost. Fine detail remained smoothed, motivating a capacity test.

v3: change capacity

Increasing the vector to 256 improved validation MSE to 0.002599 and SSIM to 0.6738. More parameters bring a modest measured reconstruction gain, without establishing detection accuracy.

Revise threshold and detector budget

The original skew-adjusted fence misrepresented a multimodal score distribution. A fixed mixture-envelope rule replaced it, with source-bootstrap uncertainty. Increasing trees from 400 to 2,000 improved repeated candidate agreement. All earlier results are retained.

Independent runs reveal a limitation

Repeating initialization and source splitting gives only 0.150 and 0.294 candidate overlap with v3, and no crop is flagged in all three runs. All five reviewed v3 examples are above the 99.7th percentile of mean brightness. Retain these weaknesses rather than treating candidates as confirmed discoveries.

v7: Gradient Loss

Introduced gradient loss to penalize blurry high-frequency edges. Broke the brightness confound and flagged true structural novelty.

v8: Metadata Fusion

Injected latitude and sun angle into latent space. Proved that metadata fusion did not significantly alter the novelty distribution vs image-only.

Remaining limitations. Hidden labels prevent ranking methods by actual anomaly recall. Fine-scale reconstruction remains imperfect. Gaussian mixtures can misfit tails or absorb contamination; limited source groups widen calibration uncertainty. Similar candidates may reflect ordinary terrain or acquisition footprints. The tested compute budget is finite and does not establish a global optimum.

A full symptom, diagnosis, fix and outcome journal is provided in ENGINEERING_CHANGELOG.md. Checkpoints, per-epoch histories and exact configurations are retained. Recorded wall times include local contention or host interruptions and are not controlled GPU benchmarks.

References and provenance

Competition sources. DA PS 26'.docx; DATA_DESCRIPTION.docx; RULES AND REGULATIONS.docx; supplied dataset archive.

Wang et al. 2004, SSIM. <https://www.cns.nyu.edu/~lcv/ssim/>

Odena et al. 2016, resize-convolution. <https://distill.pub/2016/deconv-checkerboard/>

scikit-learn, Isolation Forest. <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html>

scikit-learn, Gaussian mixture model selection. https://scikit-learn.org/stable/auto_examples/mixture/plot_gmm_selection.html

Hubert and Vandervieren 2008. <https://doi.org/10.1016/j.csda.2007.11.008>

Bouman and Heskes 2025. <https://arxiv.org/abs/2501.13864>